

- 5 Several factors are associated with the risk of developing cardiovascular disease (CVD).

Obesity is one risk factor associated with CVD.

The ability of a group of people to lose weight was investigated.

This investigation considered:

- the body mass index (BMI) of the person
- the levels of amylase in their saliva
- the bacteria present in their digestive system.

- (a) The BMI of a person can be calculated using the formula:

$$\text{BMI} = \text{mass in kg} \div (\text{height in m})^2$$

- (i) Name **one** other obesity indicator.

(1)

- (ii) Which of the following is the BMI of a person who has a mass of 65 kg and a height of 165 cm?

(1)

- ☐ A 23
- ☐ B 24
- ☐ C 39
- ☐ D 40

- (iii) A person's BMI is used to put them into a weight category.

A person with a BMI of 25 and above, but below 30, is considered to be overweight but not obese.

Which describes the body mass of a person who is obese?

(1)

- ☐ A BMI > 30
- ☐ B BMI $\geq$ 30
- ☐ C BMI < 30
- ☐ D BMI $\leq$ 30

(b) People who are overweight or obese are encouraged to lose weight.

State **two** ways in which a person could lose weight.

(1)

1

2

(c) Amylase in saliva breaks down amylose into maltose.

(i) Which row of the table describes the action of amylase?

(1)

	Type of reaction	Glycosidic bond broken
☐ A	condensation	1 – 4
☐ B	condensation	1 – 6
☐ C	hydrolysis	1 – 4
☐ D	hydrolysis	1 – 6

(ii) People with low levels of amylase in their saliva were found to lose weight more easily.

Suggest why these people lost weight more easily.

(2)

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- (d) The bacteria in people's digestive systems replicate.

The investigation found that people with greater weight loss had bacteria with increased rates of replication.

- (i) Explain why this finding can be described as a positive correlation.

(2)

- (ii) Suggest why the rate of replication of the bacteria could affect weight loss.

(2)

(Total for Question 5 = 11 marks)

